

Custom Roll Creation Using Blender

Ava Farkash

Summer 2026

Contents

1	Planning Dimensions and Setting Up the Blender File	3
1.1	Procedure Assumptions	3
1.2	Defining Variables Used in this Guide	3
1.3	Calculating Project Values	3
1.4	Converting Geometry Nodes Distances to Meters	4
1.5	Create a Blender File	4
1.6	Set the Correct Units	4
1.7	Choose the Cylinder Mesh Resolution	5
2	Creating the Base Cylinder	5
2.1	Add the Cylinder	5
2.2	Adjustments	5
2.3	Add Subdivisions	6
2.4	Verify the Cylinder	7
3	Creating the Tile Plane	7
4	Building and Baking Grayscale Material	8
4.1	Creating the Procedural Grayscale	8
4.2	Connecting the Pattern for Emission Baking	9
4.3	Create the Height-map Image	9
4.4	Bake the Procedural Pattern into the Image	10
4.4.1	Change the Render Engine	10
4.4.2	Configure Baking	10
4.4.3	Bake	10
4.4.4	Inspect the Image	10
4.4.5	Troubleshooting	11
4.4.6	Save the Height-map	11
4.5	Create the Cylinder's Geometry Nodes Modifier	11
4.5.1	Read the Cylinder Position	12
5	Wrapping the Height-map	12
5.1	Add Arctan2	12
5.2	Add π	12
5.3	Divide by 2π	13
5.4	Multiply by the Horizontal Repeat Count	13
5.5	Calculate the Vertical Pattern Coordinate	13
5.6	Combine the Image Coordinates	14
5.7	Sample the Baked Height-map	14
6	Gray Value Conversion	14
6.1	Extract the Grayscale Channel	14
6.2	Center Gray at Zero Displacement	15
6.2.1	Subtract 0.5	15

6.2.2	Multiply by the Complete Relief Depth	15
7	Restricting Displacement	15
7.1	Create the Radial Displacement Direction	16
7.1.1	Create the Radial Vector	16
7.1.2	Normalize It	16
7.1.3	Scale it by the Height Value	16
7.2	Restrict Displacement to Chosen Pattern Height	16
7.2.1	Bottom Comparison	16
7.2.2	Top Comparison	16
7.2.3	Combine the Selections	17
8	Testing and Restoring	17
8.1	Complete the Node Flow	17
8.2	Test the Node System	17
8.2.1	Improve Visibility	17
8.2.2	When the Entire Cylinder Changes Size Uniformly	18
8.3	Restore the Final Pattern	18
9	Seam Inspection	18
9.1	Confirm the Pattern is Actual Geometry	19
9.1.1	Solid-view Test	19
9.1.2	Modifier Toggle Test	19
9.2	Pack the Height-map	19
10	Creating a Backup	20
11	Applying Geometry Nodes	20
12	Recalculating Normals and Checking the Mesh	20
12.1	Check the Mesh	21
12.2	Optional Triangulation	21
13	Verifying Dimensions	22
14	Exporting the File (STL)	22
14.1	Enable 3D Print Toolbox	22
14.2	Export	23
14.3	Verifying the Exported File	23

AI Disclosure: Claude AI and ChatGPT were both used in a research context to teach software (Blender), generate code (Overleaf LaTeX), and answer general questions. Generative AI (ChatGPT) created the basis for this guide alongside manual editing and LaTeXing.

1 Planning Dimensions and Setting Up the Blender File

1.1 Procedure Assumptions

- The pattern is created procedurally in the plane's material.
- The cylinder is vertical along the Z axis.
- One plane represents one repeating texture tile.
- The material is baked into a grayscale height-map.
- Geometry Nodes wraps that height-map around the cylinder.
- The cylinder's original radius corresponds to 50
- Black cuts inward, white rises outward.
- The textured region stops sharply at its boundaries; no fade is added.
- The finished Geometry Nodes result is applied and exported as an STL

1.2 Defining Variables Used in this Guide

Symbol	Definition
R	Cylinder Radius
H	Total cylinder height
P	Height of patterned region
Tx	Physical width of one repeated tile
Ty	Physical height of one repeated tile
D	Total black-to-white relief depth
S	Suggested mesh samples per tile, initially 8

Table 1: Variables and Definitions

1.3 Calculating Project Values

Circumference: $2\pi R$

Blank band at each end of cylinder (assuming you want the cylinder to have a patterned section with blank ends): $B = (H - P) \div 2$

The number of repeats around the cylinder should normally be an integer: $Horizontal\ Repeats = round(C \div Tx)$

The number of vertical repeats can be fractional: $Vertical\ Repeats = P \div Ty$

Note 1: A fractional vertical count creates a partial tile at one end of the patterned area. To use only complete vertical tiles, choose an integer repeat count and adjust P.

Note 2: When $C \div Tx$ is not an integer, choose one of these: (1) Round the repeat count and allow the tile to stretch slightly around the cylinder or (2) Change the cylinder radius to: $R = \text{Horizontal Repeats} \cdot Tx \div (2\pi)$ The second option preserves the exact physical tile width and creates a perfect integer wrap.

1.4 Converting Geometry Nodes Distances to Meters

Q: Why are we converting to metric?

A: With Blender's Unit Scale left at 1.000, Imperial units primarily affect how measurements are displayed. On the other hand, Geometry Nodes position fields and generic Math-node values are evaluated using Blender's internal meter-based values, so we need to convert.

Conversion: $Meters = Inches \cdot 0.0254$

Symbol	Meter Value
R	
H	
P	
D	

Table 2: Your Variables in Metric

1.5 Create a Blender File

1. Launch Blender
2. Click File $\rightarrow$ New $\rightarrow$ General
3. Click the default cube, press X, then delete
4. Click File $\rightarrow$ Save As
5. Save the file as something recognizable (Ex: pattern_roll_base.blend)

1.6 Set the Correct Units

1. In the right-side Properties Editor, click Scene Properties and hover over the vertical icons until the tooltip says Scene Properties.
2. Expand Units
3. Set:

Unit System: Imperial

Unit Scale: 1.000

Length: Inches

Rotation: Degrees

1.7 Choose the Cylinder Mesh Resolution

The mesh must contain enough vertices for Set Position to form the hills and valleys. Shading alone does not add geometric detail. A reasonable starting point is approximately eight mesh intervals across each repeating tile:

$$\text{Circumference vertices} \approx \text{horizontal repeats} \cdot 8$$

For the vertical direction:

$$\text{Height segments} \approx \text{ceiling}((H \div Ty) \cdot 8)$$

Increase these values when:

- The tile contains multiple small features
- Curves look polygonal
- Peaks appear clipped or blocky
- The final manufacturing process can reproduce finer detail

Decrease them when Blender becomes unmanageably small. Practical values for a 12" tall roll: Circumference vertices: 1024 and Vertical segments: 768. This is a bit up to when you judge the pattern to be sufficiently visible and exportable.

2 Creating the Base Cylinder

Blender's cylinder primitive allows the perimeter vertex count, radius, depth, and cap type to be specified during creation.

2.1 Add the Cylinder

Place the cursor at the origin:

1. Move the pointer into the 3D Viewport
2. Press Shift+S
3. Click the cursor to World Origin

Add the cylinder:

1. Press Shift+A
2. Click Mesh → Cylinder
3. Before clicking elsewhere, press F9

2.2 Adjustments

In the Adjust Last Operation panel, enter:

Vertices: calculated circumference vertex count

Radius: R with a unit suffix

Depth: H with a unit suffix

Cap Fill Type: N-gon

Generate UVs: enabled

Align: World

Location X: 0

Location Y: 0

Location Z: 0

Rotation X: 0

Rotation Y: 0

Rotation Z: 0

Example:

Vertices: 1024

Radius: 2.5in

Depth: 12in

Click outside the panel when finished.

2.3 Add Subdivisions

The cylinder initially has many vertices around its circumference but very few along its height.

1. Verify that the cylinder is selected
2. Press Tab to enter Edit Mode
3. Press Ctrl+R
4. Move the pointer over the cylinder sidewall
5. Wait for a yellow horizontal preview loop
6. Type the desired number of cuts: $Cuts = desired\ height\ segments - 1$

Example:

Desired segments: 768

Number of cuts: 767

7. Left-click once

8. Immediately right-click once to center the loops evenly
9. Wait while Blender processes the mesh
10. Press Tab to return to Object Mode

Do not repeatedly click while a dense mesh is being generated.

2.4 Verify the Cylinder

1. Press N in the 3D Viewport
2. Click the Item tab
3. Confirm:

Location: 0, 0, 0

Rotation: 0, 0, 0

Scale: 1, 1, 1

Dimensions:

$X = 2R$

$Y = 2R$

$Z = H$

If Scale is not 1, 1, 1:

1. Press Ctrl+A
2. Click Scale

Rename the cylinder:

1. Press F2
2. Enter: Roll_Procedural
3. Press Enter

For smoother display:

1. Right-click the cylinder
2. Choose Shade Auto Smooth, Shade Smooth by Angle, or the equivalent smooth-shading option in your Blender build. This affects display normals, not the physical relief.

3 Creating the Tile Plane

1. Press Shift+A
2. Click Mesh → Plane

3. Press N

4. In the Item tab, set:

Dimension X: Tx

Dimension Y: Ty

Example:

X: 0.119in

Y: 0.119in

5. Press Ctrl+A

6. Click Scale

7. Press F2

8. Rename it: Pattern_Tile_Plane

Next, confirm the plane's UV map:

1. Press Tab to enter Edit Mode

2. Press A

3. Press U

4. Click Reset

5. Press Tab to return to Object Mode

Note 3: A bake target requires a UV map and an active Image Texture or Color Attribute.

4 Building and Baking Grayscale Material

4.1 Creating the Procedural Grayscale

1. Select Pattern_Tile_Plane

2. Click the Shading workspace at the top

3. In the Shader Editor, click New

Build the desired procedural texture using nodes such as:

- Voronoi Texture
- Noise Texture
- Wave Texture
- Math nodes

- Vector Math nodes
- ColorRamp

The final pattern must represent height as grayscale:

Black = deepest valley
 50% gray = original cylinder radius
 White = highest hill

4.2 Connecting the Pattern for Emission Baking

1. Press Shift+A in the Shader Editor
2. Search for Emission
3. Add the Emission shader
4. Connect the final grayscale output to: Emission → Color
5. Set Strength to 1.000
6. Connect: Emission → Material Output → Surface

Note 4: Do not use lighting, shadows, roughness, or a bump node as the baked height source. The emitted grayscale value itself should represent height.

4.3 Create the Height-map Image

1. Move the pointer into the Shader Editor
2. Press Shift+A
3. Click Texture → Image Texture
4. Place the node in an empty area
5. Click New on the Image Texture node

Use settings such as:

Name: tile_heightmap

Width: 2048

Height: 2048

Generated Type: Blank

Float Buffer: enabled when available

A resolution of 2048×2048 is a good starting point. Use 4096×4096 when the tile contains very fine detail and the computer has enough memory.

6. Click OK
7. Leave the Image Texture node unconnected
8. Click the node once so it has the active bright outline

Note 5: The active Image Texture node is the destination for the bake.

4.4 Bake the Procedural Pattern into the Image

4.4.1 Change the Render Engine

1. In the Properties Editor, click Render Properties, the camera icon
2. Set: Render Engine: Cycles

4.4.2 Configure Baking

1. In Render Properties, expand Bake
2. Set the bake type to: Emit (Some versions may label it Emission)
3. Set:

Selected to Active: disabled

Clear Image: enabled

Margin: 16 px

4.4.3 Bake

1. Confirm only the plane is selected
2. Confirm the unconnected tile_heightmap Image Texture node is active
3. Click Bake
4. Wait until Blender finishes

Note 6: Cycles baking requires a UV-mapped mesh and uses the active Image Texture node as its target.

4.4.4 Inspect the Image

1. Change one editor to an Image Editor, or use the existing Image Editor
2. Select tile_heightmap from its image drop-down
3. Move the pointer over the Image Editor
4. Press Home

The image should contain visible black, gray, and white variation.

4.4.5 Troubleshooting

If the baked image is completely black, check:

- The procedural output is connected to Emission Color
- Emission is connected to Material Output Surface
- The correct Image Texture node is active
- The plane has a UV map
- Cycles is selected
- The bake type is Emit/Emission
- The target image was created before baking

A completely black image makes every cylinder sample identical and can shrink the entire patterned section when the displacement math subtracts 0.5.

4.4.6 Save the Height-map

1. Move the pointer over the Image Editor
2. Click Image → Save As
3. Enter a filename: tile_heightmap_16bit.png
4. Set: File Format: PNG Color: BW or RGB Color Depth: 16
5. Click Save As Image (Avoid JPEG because compression artifacts can become unwanted physical surface detail)

4.5 Create the Cylinder's Geometry Nodes Modifier

1. Click the Geometry Nodes workspace
2. Select Roll_Procedural in the Outliner
3. Click New in the Geometry Nodes editor

You should see:

Group Input

Group Output

Add a Set Position:

1. Press Shift+A in the Geometry Nodes editor
2. Search for Set Position
3. Place it between Group Input and Group Output

4. Connect: Group Input Geometry $\rightarrow$ Set Position Geometry $\rightarrow$ Group Output Geometry

Note 7: Set Position physically translates selected mesh points through its Position or Offset inputs.

4.5.1 Read the Cylinder Position

1. Press Shift+A
2. Search for Position
3. Add the Position node
4. Press Shift+A again
5. Search for Separate XYZ
6. Add it
7. Connect: Position $\rightarrow$ Separate XYZ Vector

The outputs now represent the cylinder point coordinates: X Y Z

5 Wrapping the Height-map

5.1 Add Arctan2

1. Press Shift+A
2. Search for Math
3. Add a Math node
4. Change its operation to Arctan2
5. Connect: Separate XYZ Y $\rightarrow$ Arctan2 top input Separate XYZ X $\rightarrow$ Arctan2 bottom input

5.2 Add π

1. Add another Math node
2. Set it to Add
3. Connect the Arctan2 output to its first input
4. Enter in the second input: pi

5.3 Divide by 2π

1. Add another Math node
2. Set it to Divide
3. Connect the Add output to its first input
4. Enter in the second input: 2π

This produces a value running approximately from 0 to 1 around the cylinder.

5.4 Multiply by the Horizontal Repeat Count

1. Add another Math node
2. Set it to Multiply
3. Connect the Divide output to its first input
4. Enter the integer horizontal repeat count in the other input

The completed chain is: Position X and Y $\rightarrow$ Arctan2(Y, X) $\rightarrow$ Add π $\rightarrow$ Divide by 2π $\rightarrow$ Multiply by horizontal repeats

5.5 Calculate the Vertical Pattern Coordinate

1. Press Shift+A
2. Add a Map Range node
3. Connect: Separate XYZ Z $\rightarrow$ Map Range Value

Set:

Data Type: Float

Interpolation: Linear

From Min: $-P/2$ in meters

From Max: $+P/2$ in meters

To Min: 0

To Max: vertical repeat count

Clamp: enabled

Example:

From Min: -0.127

From Max: 0.127

To Min: 0

To Max: 84.033613

Map Range remaps values between specified source and target ranges and can clamp values to the target range.

5.6 Combine the Image Coordinates

1. Press Shift+A
2. Search for Combine XYZ
3. Add it
4. Connect: Horizontal Multiply output → Combine XYZ X
Vertical Map Range Result → Combine XYZ Y
5. Leave Z at 0

5.7 Sample the Baked Height-map

1. Press Shift+A
2. Search for Image Texture
3. Add the Geometry Nodes Image Texture node
4. Select the baked tile_heightmap image
5. Connect: Combine XYZ Vector → Image Texture Vector

Set:

Interpolation: Linear

Extension: Repeat

The Geometry Nodes Image Texture node samples the selected image using the supplied Vector coordinates. Set the image's color space to Non-Color when that control is available in Image Properties or the Image Editor.

6 Gray Value Conversion

6.1 Extract the Grayscale Channel

1. Press Shift+A
2. Search for Separate Color
3. Add it
4. Leave the mode at RGB

5. Connect: Image Texture Color $\rightarrow$ Separate Color Color

Use the Red output. A true grayscale image has equivalent red, green, and blue values.

6.2 Center Gray at Zero Displacement

Do not use an asymmetrically edited Map Range for this step. Changing only its To Min value can shift most of the cylinder inward or outward.

6.2.1 Subtract 0.5

1. Add a Math node
2. Set it to Subtract
3. Connect: Separate Color Red $\rightarrow$ Subtract top input
4. Set the lower input to: 0.500

This produces:

Black: -0.5

50% gray: 0

White: +0.5

6.2.2 Multiply by the Complete Relief Depth

1. Add another Math node
2. Set it to Multiply
3. Connect the Subtract output to its first input
4. Enter D converted to meters in the second input

Example:

$$0.016 \text{ in} \times 0.0254 = 0.0004064 \text{ m}$$

Because the centered grayscale range is -0.5 through +0.5, this creates:

Black: $-D/2$

Gray: 0

White: $+D/2$

7 Restricting Displacement

Using a calculated radial vector ensures that points move directly toward or away from the cylinder axis.

7.1 Create the Radial Displacement Direction

7.1.1 Create the Radial Vector

1. Add a second Combine XYZ node
2. Connect: Separate XYZ X $\rightarrow$ second Combine XYZ X
Separate XYZ Y $\rightarrow$ second Combine XYZ Y
3. Leave Z at 0

7.1.2 Normalize It

1. Add a Vector Math node
2. Set its operation to Normalize
3. Connect: Second Combine XYZ $\rightarrow$ Normalize

7.1.3 Scale it by the Height Value

1. Add another Vector Math node
2. Set it to scale
3. Connect: Normalize Vector $\rightarrow$ Vector Math Scale node Vector
Relief Multiply $\rightarrow$ Vector Math Scale node Scale
4. Connect its output to: Set Position $\rightarrow$ Offset

The image now controls real vertex displacement.

7.2 Restrict Displacement to Chosen Pattern Height

7.2.1 Bottom Comparison

1. Add a Compare node
2. Set: Data Type: Float
Operation: Greater Than
3. Connect: Separate XYZ Z $\rightarrow$ A
4. Set B to: $-P/2$ in meters

7.2.2 Top Comparison

1. Duplicate the Compare node with Shift+D
2. Change its operation to Less Than
3. Connect Z to A

4. Set B to: $+P/2$ in meters

7.2.3 Combine the Selections

1. Add a Boolean Math node
2. Set it to AND
3. Connect both Compare outputs into it
4. Connect: Boolean Math AND $\rightarrow$ Set Position Selection

This leaves the cylinder's end bands unchanged and applies displacement only within the centered patterned region.

8 Testing and Restoring

8.1 Complete the Node Flow

The geometry path is: Group Input $\rightarrow$ Set Position $\rightarrow$ Group Output

The coordinate path is: Position $\rightarrow$ Separate XYZ

X and Y: Arctan2 $\rightarrow$ Add π $\rightarrow$ Divide by 2π $\rightarrow$ Multiply by horizontal repeats $\rightarrow$ Combine XYZ X

Z: Map Range $\rightarrow$ Combine XYZ Y

Combine XYZ $\rightarrow$ Image Texture

The relief path is: Image Texture Color $\rightarrow$ Separate Color Red $\rightarrow$ Subtract 0.5 $\rightarrow$ Multiply by full relief depth $\rightarrow$ Vector Math Scale $\rightarrow$ Set Position Offset

The radial direction path is: Position X and Y $\rightarrow$ Combine XYZ with Z=0 $\rightarrow$ Normalize $\rightarrow$ Vector Math Scale Vector

The selection path is: $Z > -P/2$ AND $Z < +P/2$ $\rightarrow$ Set Position Selection

8.2 Test the Node System

Before using the final fine pattern, temporarily make it obvious. Set:

Horizontal repeats: 8

Vertical repeats: 8

Relief depth: 0.100 in = 0.00254 m

8.2.1 Improve Visibility

1. Move into the 3D Viewport

2. Select Solid Viewport Shading
3. Click the dropdown beside the shading icons
4. Enable Cavity
5. Select Both when available
6. Orbit and zoom close to the cylinder. A large, obvious physical pattern should appear.

8.2.2 When the Entire Cylinder Changes Size Uniformly

Check these items:

1. The baked image is not completely black
2. The Image Texture node contains the correct image
3. The relief chain contains: Red $\rightarrow$ Subtract 0.5 $\rightarrow$ Multiply
4. The Multiply depth is expressed in meters
5. The radial Vector Math output is connected to Offset, not Position
6. The pattern-selection AND output is connected to Selection

8.3 Restore the Final Pattern

After the test succeeds, restore:

Horizontal repeats = calculated integer wrap count

Vertical repeats = $P \div T_y$

Relief depth = final D converted to meters

Example:

Horizontal repeats: 132

Vertical repeats: 84.033613

Relief depth: 0.0004064

9 Seam Inspection

1. Temporarily return the repeat counts to 8 if the fine pattern is difficult to inspect
2. Temporarily increase depth to 0.00254
3. Use View $\rightarrow$ Viewpoint $\rightarrow$ Left, Right, Front, or Back to locate the vertical seam
4. Zoom close to it
5. Orbit slightly

Look for:

- A vertical ridge
- A vertical groove
- A sudden change in height
- Repeating horizontal or vertical tile-edge lines

A seamless result requires:

- An integer number of tiles around the circumference
- Matching left and right edges in the baked height-map
- Preferably matching top and bottom tile edges when it repeats vertically

After checking, restore the real repeat counts and depth.

9.1 Confirm the Pattern is Actual Geometry

9.1.1 Solid-view Test

1. Use Solid Viewport Shading
2. Zoom in and orbit
3. Look at the cylinder silhouette

A material color or bump map cannot alter the silhouette. Displaced geometry can.

9.1.2 Modifier Toggle Test

1. Select the cylinder
2. Open Properties → Modifiers, the wrench icon
3. Find the Geometry Nodes modifier
4. Click its viewport monitor icon to disable it; the cylinder should become smooth
5. Turn the monitor icon back on; the physical pattern should return

Set Position changes the evaluated point positions, so the pattern is geometric rather than merely colored.

9.2 Pack the Height-map

1. Click File → External Data → Pack Resources
2. Click File → Save

This keeps the image with the .blend file if the external PNG is moved.

10 Creating a Backup

1. Select the cylinder
2. Press F2
3. Rename it: Roll_Procedural_BACKUP
4. Press Shift+D
5. Immediately right-click so the duplicate does not move
6. Press F2
7. Rename the duplicate: Roll_Final_Applied
8. In the Outliner, click the eye icon beside Roll_Procedural_BACKUP to hide it
9. Keep Roll_Final_Applied selected
10. Save the file under a new name: pattern_roll_pre_export.blend

11 Applying Geometry Nodes

1. Confirm Roll_Final_Applied is selected
2. Confirm the mode selector says Object Mode
3. Open Properties → Modifiers
4. Find the Geometry Nodes modifier
5. Open its small drop-down menu
6. Click Apply
7. Wait while Blender generates the final dense mesh
8. Save the file

The pattern should remain visible after the modifier disappears.

12 Recalculating Normals and Checking the Mesh

1. Press Tab to enter Edit Mode
2. Press A
3. Press Z
4. Click Wireframe (The mesh edges should physically follow the surface relief)
5. Press Z again.
6. Click Solid

7. Press Shift+N
8. Choose Recalculate Outside
9. Press Tab to return to Object Mode
10. Save

12.1 Check the Mesh

1. Select Roll_Final_Applied
2. Move the pointer into the 3D Viewport
3. Press N
4. Click the 3D-Print tab
5. Expand Checks
6. Click Check All

Pay particular attention to:

- Non-Manifold
- Bad Contiguous
- Intersections
- Degenerate

These should ideally report zero. A manufacturing mesh should form a closed, watertight surface without loose edges or holes.

12.2 Optional Triangulation

An STL file will ultimately store triangles. When the toolbox reports distorted non-planar quads:

1. Open Properties → Modifiers
2. Click Add Modifier
3. Search for Triangulate
4. Add it
5. Apply it
6. Run Check All again

A nonzero distorted count alone does not always mean the solid is open, but non-manifold edges, intersections, or degenerate faces should be investigated before export.

13 Verifying Dimensions

1. Select the applied roll
2. Press N
3. Open the Item tab
4. Verify the total Z dimension equals H

For the middle-gray neutral-radius method:

$$\text{Minimum radius} = R - D/2$$

$$\text{Maximum radius} = R + D/2$$

$$\text{Minimum diameter} = 2R - D$$

$$\text{Maximum diameter} = 2R + D$$

Example:

$$R = 2.5 \text{ in}$$

$$D = 0.016 \text{ in}$$

$$\text{Minimum diameter} = 4.984 \text{ in}$$

$$\text{Maximum diameter} = 5.016 \text{ in}$$

$$\text{Height} = 12 \text{ in}$$

The X and Y dimensions may differ slightly if the highest white areas do not occur at the exact X- and Y-axis positions.

14 Exporting the File (STL)

14.1 Enable 3D Print Toolbox

If you want to 3D print the custom roll you are about to create, you have to enable the 3D Print Toolbox in Blender. This Toolbox provides checks for problems that can prevent a mesh from being watertight or printable. To enable it:

1. Click Edit → Preferences
2. Click Get Extensions
3. Search: 3D Print Toolbox
4. Click Install
5. Open Add-ons in Preferences
6. Confirm the toolbox is enabled

7. Close Preferences

14.2 Export

STL is commonly used for CAD and 3D-printing workflows, and Blender's importer/exporter provides scaling and scene-unit controls. Because STL does not reliably carry a unit definition, export values that downstream manufacturing software will interpret as millimeters.

1. Select only Roll_Final_Applied
2. Click File → Export → STL (.stl)
3. Choose the destination folder
4. Enter a descriptive filename: pattern_roll_final_mm.stl
5. In the export options, set:

Selection Only: enabled

Apply Modifiers: enabled

Scale or Global Scale: 1000

Scene Unit / Apply Unit: disabled

The 1000 scale converts meter-based coordinate values into millimeter-valued STL coordinates. Do not enable another unit-conversion option at the same time, or the model may be scaled twice.

6. Leave the file in binary format unless ASCII STL is specifically required
7. Click Export STL
8. Wait for Blender to finish

14.3 Verifying the Exported File

Open the STL in the actual slicer, CAM program, or inspection software. Confirm:

Height = H converted to millimeters

Nominal diameter = 2R converted to millimeters

Maximum and minimum diameter include the chosen relief.

For the 12 inch Roll example:

Height: 304.8 mm

Neutral diameter: 127.0 mm

Maximum diameter: 127.4064 mm

Minimum diameter: 126.5936 mm

Inspect:

- The wrap seam
- The top and bottom blank bands
- The patterned height
- The relief direction
- The silhouette
- The STL's reported scale
- Any unexpectedly missing or filled areas

Reimporting the STL into a blank Blender file is also a useful final scale and geometry check.